School of Computer Science Engineering and Information Systems (SCORE)

Internship Project done at:

Envision Financial Systems (India) Private Limited

Project Title

Combination of wires – Consolidate Money Transactions

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Acknowledgement

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Keywords

- Financial Transaction Consolidation: Combining multiple transactions into a single payment to reduce processing fees.
- Wire Transfers: Electronic movement of funds between financial institutions, often incurring fixed fees.
- Synthetic Data Generation: Automatically creating realistic transaction data to simulate daily business operations.
- Python Automation: Using Python scripts to streamline data generation, reporting, and email delivery.
- CSV Report Generation: Saving structured daily transaction and summary reports in a readable, spreadsheet-friendly format.
- Email Automation: Sending reports securely via Gmail’s SMTP service using App Passwords.
- Batch Processing: Executing multiple scripts in sequence using a Windows batch file.
- Task Scheduling: Running the automation pipeline daily through Windows Task Scheduler without manual input.
- Operational Efficiency: Reducing manual effort, errors, and costs through automation.

Organization Profile

Envision Financial Systems (India) Private Limited is the Indian subsidiary of Envision Financial Systems Inc., a U.S.-based fintech company founded in 1994. The parent company is known for providing robust shareholder accounting and investment management solutions to the global financial services industry. Envision's clients include mutual fund companies, 529 college savings plan administrators, asset managers, transfer agents, and financial recordkeepers. With a strong presence in the U.S. market, Envision delivers reliable, scalable software platforms that enable firms to process large volumes of financial transactions with accuracy and compliance.

The Indian operations, headquartered in Hyderabad, play a pivotal role in Envision's global technology strategy. The team in India is responsible for a wide range of activities including core product development, software customization, data migration, quality assurance, and ongoing technical support. The Hyderabad center functions as a collaborative R&D environment, contributing to the modernization of financial systems, integration of API-driven services, and enhancement of user interface experiences for clients.

Envision's software offerings are designed to meet stringent regulatory requirements from organizations like the U.S. Securities and Exchange Commission (SEC) and FINRA. Their solutions support operational efficiency, transaction transparency, and business continuity — all crucial for modern financial operations. The company also emphasizes cybersecurity, automation, and cloud-ready architecture, ensuring their platforms stay ahead of emerging threats and technological shifts.

With a commitment to excellence, innovation, and employee development, Envision Financial Systems (India) Pvt. Ltd. provides a dynamic work environment where professionals contribute to real-world financial solutions used across global markets. The India team continues to be a critical part of Envision's mission to help financial institutions reduce risk, improve efficiency, and deliver better client experiences through advanced software technology.

Problem Definition

In many organizations, employees who travel for work incur multiple daily expenses related to transportation, accommodation, meals, and other official activities. These expenses are typically reimbursed by the company, often through a third-party facilitator who manages and processes the transactions. However, each reimbursement transaction usually carries a fixed processing or transfer fee. When an employee submits multiple small transactions in a single day, sending each one individually leads to multiple fees for the same customer — resulting in unnecessary overhead costs.

The absence of a system that consolidates these daily transactions into a single summary per customer increases operational inefficiency and financial waste. Manual processing of such data is also time-consuming, error-prone, and not scalable. There is a clear need for an automated solution that can generate daily transaction data, consolidate it by customer, and deliver a summarized report for streamlined reimbursement — all without human intervention.

This project addresses this problem by simulating a real-world transaction environment and providing an end-to-end automated solution that includes data generation, customer-wise summarization, and secure email-based reporting. The goal is to minimize transaction charges and administrative effort while improving the speed and accuracy of financial reconciliation

Introduction

In the banking and financial world, a wire transfer refers to the electronic movement of funds from one individual or entity to another across financial institutions. The term "wire" originates from the historical use of telegraph wires to communicate money transfer instructions — although today's systems are entirely digital. Wire transfers are commonly used for high-value, urgent, or international transactions, where security and traceability are essential. Unlike traditional cheque-based or mobile methods, wire transfers are direct bank-to-bank transactions and typically clear within the same or next business day.

Some of the key characteristics of wire transfers include:

- Electronic and Direct: Funds are moved digitally from one bank to another.
- Secure: They are routed through formal networks like SWIFT (international), RTGS (India), or Fedwire (U.S.).
- Fast: Domestic wires often complete within the same day, while international wires may take 1–3 business days.
- Costly: They carry a fixed fee per transaction, making frequent small transfers expensive.
- Irreversible: Once processed, they usually cannot be canceled, increasing transaction reliability.

In this context, the concept of the "Combination of Wires" becomes relevant. It refers to the practice of merging multiple small transactions into a single consolidated transfer. This is particularly useful when an entity needs to make several payments to the same recipient — such as in corporate reimbursements or recurring expense settlements. Instead of executing and paying for each transaction individually, all are grouped and processed as one, significantly reducing the total transaction cost.

For example, consider an employee who makes 10 business purchases in one day. If a third-party financial handler sends each of these as separate wire transfers, it would result in 10 transaction fees. But by combining them into a single payment representing the total amount owed, only one transaction fee is incurred. Internally, all transaction details are still recorded for accounting and auditing purposes, but externally, only one consolidated wire is executed.

This project, titled "Combination of Wires – Consolidate Money Transactions," is designed around this very principle. The system simulates daily financial activity across a group of employees (represented as customers), aggregates all daily transactions per customer into a summarized report, and automates the email delivery of this report. This allows the third-party facilitator to process just one reimbursement per customer per day, instead of many, resulting in:

- Reduced transaction fees
- Simplified reconciliation and reporting
- Improved operational efficiency
- Clean, auditable daily financial records

Through automated scripting, scheduling, and secure email communication, the project effectively models a real-world solution for organizations looking to reduce overhead in transaction handling and financial operations.

Modules and Workflow

This project is divided into three core modules, each responsible for a key part of the automated financial transaction handling process. Together, these modules create a seamless daily workflow that begins with data generation and ends with automated communication — all without requiring any manual intervention.

1. Dataset Generation:
   This module is responsible for simulating synthetic financial transaction data. Each day, a Python script generates a dataset of exactly 5,000 transactions. These transactions are assigned randomly to between 200 and 300 unique customer IDs, simulating employees of a company who incur daily expenses during work-related activities. About 95% of these customers are assigned multiple transactions in a single day, reflecting real-world spending patterns, while the remaining 5% have only one. Each record includes key fields like Transaction ID, Customer ID, Client ID, Amount, Date, Mode (e.g., UPI, NEFT, RTGS, Wire), and Status.

2. Daily Report Generator:
   Once the dataset is created, a second Python script processes it to generate a summary report. It groups the data by Customer_ID and Date, calculates the total amount spent per customer, and counts the number of transactions. This consolidated report allows a third-party financial handler to reimburse customers in one combined payment per day rather than separate payments for every transaction. The report is saved as a CSV file and

named according to the current date, maintaining a consistent and traceable log of daily summaries.

3. Email Sender:
   The third module is responsible for automating the delivery of the daily report via email. Using Gmail's SMTP server, the system sends the CSV report as an email attachment to a designated recipient (such as the company's finance team). The script uses Python's smtplib and email.message libraries and is configured with a secure Gmail App Password. This step ensures that the recipient receives an up-to-date summary of all customer transactions every day, enabling faster approvals, reconciliations, and reimbursements.

These three modules together form the backbone of the "Combination of Wires" project, transforming what would otherwise be a repetitive and error-prone manual task into an efficient, automated pipeline.

Technologies Used

This project leverages a combination of scripting, scheduling, and secure communication technologies to simulate, process, and deliver financial transaction data effectively:

- Python:
  Python serves as the core language for the entire system. All three scripts — dataset generation, report creation, and email automation — are implemented in Python due to its simplicity, readability, and robust library support. Libraries like pandas make it easy to manage tabular data, while built-in modules like smtplib and email.message enable email-based communication.

- Gmail App Passwords:
  To enhance email security, the project uses Gmail App Passwords instead of the main Gmail password. This prevents unauthorized access and ensures compliance with two-factor authentication policies. It allows the Python script to authenticate and send emails through Gmail's SMTP server without compromising the user's primary credentials.

- Windows Batch File (.bat):
  A batch file is used to execute all three Python scripts sequentially with a single command. This ensures that the tasks run in the correct order each day — first generating the data, then creating the report, and finally emailing it.

- Windows Task Scheduler:
  Task Scheduler is configured to run the batch file at a specific time every day (e.g., 8:00 AM). It ensures that the entire workflow operates automatically in the background. Even if the user is not present or logged in, the scheduled task ensures the dataset is generated, the report is built, and the email is sent without any manual trigger.

Together, these technologies form a cohesive system that is secure, scalable, and easy to maintain. The project not only addresses a real-world financial inefficiency but also showcases how modern scripting and automation tools can solve operational problems effectively.

Automation Setup

Automation is implemented using a Windows batch file that executes all three Python scripts—dataset generation, report creation, and email sending—in sequence. This .bat file is scheduled to run automatically every day using Windows Task Scheduler at a specific time. Once configured, the system handles the full workflow daily without requiring any manual input. The dataset and report are saved with the current date in their filenames, and the email module ensures the report is sent out on the same day. This setup ensures consistent, reliable, and repeatable daily execution of all tasks.

Outputs

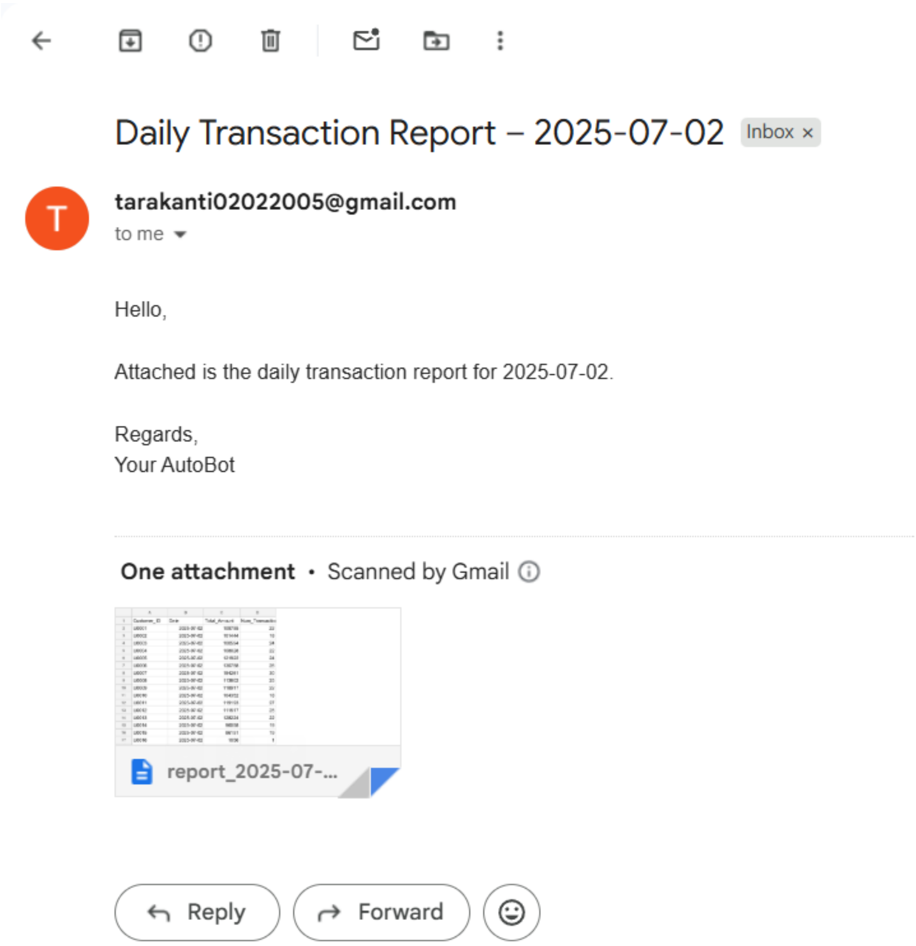
Sample Transaction Dataset

	A	B	C	D	E	F	G
1	Transaction_ID	Client_ID	Customer_ID	Amount	Date	Mode	Status
2	T000001	C001	U0196	7685	02-07-2025	NEFT	Success
3	T000002	C001	U0111	3683	02-07-2025	RTGS	Success
4	T000003	C001	U0058	9095	02-07-2025	RTGS	Success
5	T000004	C001	U0098	5253	02-07-2025	UPI	Success
6	T000005	C001	U0091	3491	02-07-2025	Wire	Success
7	T000006	C001	U0122	4900	02-07-2025	RTGS	Success
8	T000007	C001	U0143	2534	02-07-2025	NEFT	Success
9	T000008	C001	U0016	1056	02-07-2025	Wire	Success
10	T000009	C001	U0065	7976	02-07-2025	RTGS	Success
11	T000010	C001	U0020	1814	02-07-2025	RTGS	Success
12	T000011	C001	U0053	3951	02-07-2025	UPI	Success
13	T000012	C001	U0062	277	02-07-2025	UPI	Success
14	T000013	C001	U0069	8904	02-07-2025	UPI	Success
15	T000014	C001	U0090	8977	02-07-2025	NEFT	Success
16	T000015	C001	U0201	2143	02-07-2025	UPI	Success

Sample Daily Report

	A	B	C	D
1	Customer_ID	Date	Total_Amount	Num_Transactions
2	U0001	02-07-2025	109755	22
3	U0002	02-07-2025	101444	18
4	U0003	02-07-2025	100594	24
5	U0004	02-07-2025	108028	22
6	U0005	02-07-2025	121922	24
7	U0006	02-07-2025	130758	26
8	U0007	02-07-2025	154261	30
9	U0008	02-07-2025	113802	25
10	U0009	02-07-2025	118917	22
11	U0010	02-07-2025	104352	18

Email Sent



Conclusion and Future Enhancements

This project successfully demonstrates an automated system that simulates daily financial transactions, consolidates customer-wise reports, and delivers them via email — all without manual intervention. It effectively addresses the real-world problem of reducing transaction costs by combining multiple expenses per customer into a single daily summary. The modular design, use of reliable technologies, and daily scheduling make it efficient and scalable. In the future, the system can be enhanced by supporting multiple clients, integrating with real-time databases for persistent storage, and developing a visual dashboard for live monitoring using tools like Streamlit. Security measures such as file encryption and authentication can also be added to make the system production-ready and suitable for enterprise-level deployment.